Counterexamples to Two Galerkin Projection Lemmas in arXiv:1610.04776

Candidate counterexample packet for human review

27 August 2026

Status. Candidate counterexample, likely valid. The packet refutes Theorems 6.2 and 6.3 as stated in the source paper. It identifies a proof gap in the Galerkin derivation of the principal equilibrium theorems, but does not claim that those equilibrium theorems are false and does not answer the paper's final saturation-necessity question.

1 Source and claims under review

The source is M. Ali Khan and Nobusumi Sagara, "Fatou's Lemma, Galerkin Approximations and the Existence of Walrasian Equilibria in Infinite Dimensions," arXiv:1610.04776v2 (2017) [1].

On printed page 30, Theorem 6.2 states that if E is an ordered locally convex Hausdorff space and P is a continuous projection of E onto a closed vector subspace V, then $P : E \to V$ is positive. On printed pages 31–32, Theorem 6.3(i) states that, for any nested dense Galerkin scheme (V_n) in a separable Banach space and any continuous projections $P_n : E \to V_n$, every $(P_n x)$ has a subsequence converging weakly to x. Part (ii) asserts the weak-star analogue for projections Q_n on E^* (the displayed occurrence of $Q_n x$ is naturally read as $Q_n p$, as in its proof).

Let E be a (real) vector space which is endowed with an order structure defined by a reflexive, transitive, and anti-symmetric binary relation $\geq$. Then $(E, \geq)$ is called an *ordered vector space* if the following conditions are satisfied: (i) $x \geq y$ implies $x + z \geq y + z$ for every $x, y, z \in E$; (ii) $x \geq y$ implies $\alpha x \geq \alpha y$ for every $x, y \in E$ and $\alpha > 0$. The subset $E_+ = \{x \in E \mid x \geq 0\}$ is called a *positive cone* of an ordered vector space E. A subset C of a vector space E is called a *proper cone* if $C + C \subset C$, $\alpha C \subset C$ for every $\alpha > 0$, and $C \cap (-C) = \{0\}$. Every positive cone of an ordered vector space is a proper cone. Conversely, every proper cone C of a vector space E defines, by virtue of $x \geq y \Leftrightarrow y - x \in C$, an order under which E is an ordered vector with positive cone C. If V is a vector subspace of an order vector space E, then the induced ordering on V is determined by the proper cone $V \cap E_+$, and hence, $V_+ = V \cap E_+$ is a positive cone of V.

As a convention, by an *ordered lcHs*, we always assume that its positive cone is closed (see [77, V.4.(LTO)]). Let E be an ordered lcHs. A continuous linear functional $x^* \in E^*$ is said to be *positive* if $\langle x, x^* \rangle \geq 0$ for every $x \in E_+$. Denote by E_+^* the set of continuous linear functionals on E. Let E and F be ordered vector spaces. A linear operator $u : E \to F$ is said to be *positive* if $u(E_+) \subset u(F_+)$.

The following observation is also useful in Section 7.

Theorem 6.2. *Let E be an ordered lcHs and P be a continuous projection of E onto a closed vector subspace V. Then $P : E \to V$ is a positive linear operator.*

Proof. Choose any $x \in E_+$. If $Px \notin V_+$, then by the separation theorem (see [72, Theorem 3.5]), there exists $x^* \in E^*$ such that $\langle Px, x^* \rangle = 1$ and $\langle y, x^* \rangle = 0$ for every $y \in V$. Let $x = x_1 \oplus x_2$ be the direct sum decomposition of x into $x_1 \in V$ and $x_2 \in \mathcal{R}(I - P)$. Since $\langle x_1, x^* \rangle = 0$ and $\langle x_2, P^* x^* \rangle = 0$, we have

$$1 = \langle Px, x^* \rangle = \langle x_1 + x_2, P^* x^* \rangle = \langle x_1, P^* x^* \rangle = \langle Px_1, x^* \rangle = \langle x_1, x^* \rangle = 0,$$

a contradiction. Therefore, $Px \in V_+$. $\square$

6.2 Galerkin Approximations

> A quite natural idea when considering an infinite dimensional (variational) problem is to approximate it by finite dimensional problems. This has important consequences both from theoretical (existence, etc.) and the numerical point of view. [...] We stress the fact that this type of finite dimensional approximation method is

30

Figure 1: Printed page 30 of the source, containing Theorem 6.2 and its proof.

very flexible and can be applied [...] to a large number of linear or nonlinear problems. [4, p. 71].

Definition 6.1. A *Galerkin approximation scheme* of a lcHs E is a sequence $\{V^n\}_{n\in\mathbb{N}}$ of finite-dimensional subspaces of E such that for every $x \in E$, there exists a sequence $\{x_n\}_{n\in\mathbb{N}}$ with $x_n \in V^n$ for each $n \in \mathbb{N}$ and $x_n \to x$ in the locally convex topology of E.

The following result is fundamental.

Proposition 6.1. *Let $(E, \|\cdot\|)$ be a separable Banach space.*

(i) *There exists a Galerkin approximation scheme $\{V^n\}_{n\in\mathbb{N}}$ of $(E, \|\cdot\|)$ such that $V^1 \subset V^2 \subset \cdots$ and $\overline{\bigcup_{n\in\mathbb{N}} V^n}^{\|\cdot\|} = E$.*

(ii) *There exists a Galerkin approximation scheme $\{W^n\}_{n\in\mathbb{N}}$ of (E^*, w^*) such that $W^1 \subset W^2 \subset \cdots$ and $\overline{\bigcup_{n\in\mathbb{N}} W^n}^{w^*} = E^*$.*

Proof. (i): For the construction of a Galerkin approximation scheme, just take a countable dense subset $\{x_i\}_{i\in\mathbb{N}}$ of E and let V^n be the linear span of the finite set $\{x_1, \dots, x_n\}$. See [4, Proposition 3.1.1] for details.

(ii): Let S^* be the closed unit ball in E^*. Since E^* is separable with respect to the weak* topology because of the separability of E (see [77, IV.1.7]), there is a countable set $\{q_i\}_{i\in\mathbb{N}}$ in S^* such that $\overline{\{q_i\}_{i\in\mathbb{N}}}^{w^*} = S^*$. Let W^n be the linear span of the finite set $\{q_1, \dots, q_n\}$. By construction, $W^1 \subset W^2 \subset \cdots$ and $\overline{\bigcup_{n\in\mathbb{N}} W^n}^{w^*} = E^*$. Let $p \in E^* \setminus \{0\}$ be arbitrarily fixed. We must construct $p_n \in W^n$ such that $p_n \to p$ weakly*. By virtue of normalization, without loss of generality we may assume that $p \in S^*$. Since S^* is metrziable with respect to the weak* topology in view of the separability of E (see [27, Theorem V.3.1]), for a compatible metric δ on S^* with the weak* topology, there exists a mapping $k \mapsto n(k)$ from $\mathbb{N}$ into $\mathbb{N}$ such that $\delta(p, q_{n(k)}) < 1/k$ for each $k \in \mathbb{N}$. Since $q_{n(k)} \in W^{n(k)} \subset W^n$ for each $k, n \in \mathbb{N}$ with $n \ge n(k)$, we obtain $\operatorname{dist}(p, W^n) \le \operatorname{dist}(p, W^{n(k)}) < 1/k$ for each $n \ge n(k)$, and hence, $\operatorname{dist}(p, W^n) \to 0$ as $n \to \infty$, which guarantees that there exists $p_n \in W^n$ such that $\delta(p, p_n) \to 0$. □

Theorem 6.3. *Let $(E, \|\cdot\|)$ be a separable Banach space.*

(i) *If $\{V^n\}_{n\in\mathbb{N}}$ is a Galerkin approximation scheme of $(E, \|\cdot\|)$ such that $V^1 \subset V^2 \subset \cdots$ and $\overline{\bigcup_{n\in\mathbb{N}} V^n}^{\|\cdot\|} = E$, and P_n is a continuous projection of E onto V^n, then for every $x \in E$ the sequence $\{P_n x\}_{n\in\mathbb{N}}$ contains a subsequence converging weakly to x.*

31

Figure 2: Printed page 31 of the source, containing Theorem 6.3(i).

(ii) *If* $\{W^n\}_{n\in\mathbb{N}}$ *is a Galerkin approximation scheme of* (E^*, w^*) *such that* $W^1 \subset W^2 \subset \cdots$ *and* $\overline{\bigcup_{n\in\mathbb{N}} W^n}^{w^*} = E^*$, *and* Q_n *is a continuous projection of* E^* *onto* W^n, *then for every* $p \in E^*$ *the sequence* $\{Q_n x\}_{n\in\mathbb{N}}$ *contains a subsequence converging weakly* to* p.

Proof. (i): Let $x \in E$ be arbitrarily fixed. It suffices to show that there is a subsequence $\{P_m x\}_{m\in\mathbb{N}}$ of $\{P_n x\}_{n\in\mathbb{N}}$ such that $\langle p, P_m x\rangle \to \langle p, x\rangle$ for every $p \in E^*$. By Proposition 6.1(i), there exists $x_n \in V^n$ such that $x_n \to x$. Choose any $p \in E^*$. Since $\{(I-P_n^*)p\}_{n\in\mathbb{N}}$ is a bounded sequence in E^* in view of $\mathcal{R}(I-P_{n+1}^*) \subset \mathcal{R}(I-P_n^*) \subset \cdots$ for each $n \in \mathbb{N}$ and $\{x_i\}_{i\in\mathbb{N}}$ is a bounded sequence in E, the numerical double sequence $\{\langle (I-P_n^*)p, x_i\rangle\}_{(i,n)\in\mathbb{N}^2}$ contains a convergent subsequence $\{\langle (I-P_m^*)p, x_j\rangle\}_{(j,m)\in\mathbb{N}^2}$. It follows from $I - P_m^* = (V^m)^\perp$ and $x_j \in V^j$ that $\langle (I-P_m^*)p, x_j\rangle = 0$ for each $j, m \in \mathbb{N}$ with $j \le m$. Hence,

$$\begin{aligned}\lim_{m\to\infty}\sup_{j\in\mathbb{N}} |\langle (I-P_m^*)p, x_j\rangle| &= \lim_{m\to\infty}\sup_{j>m} |\langle (I-P_m^*)p, x_j\rangle| \\ &= \lim_{j,m\to\infty} |\langle (I-P_m^*)p, x_j\rangle|.\end{aligned}$$

This means that the convergence $\lim_m \langle (I-P_m^*)p, x_j\rangle = 0$ is uniform in j. We thus obtain

$$\begin{aligned}\lim_{m\to\infty}\langle p, x - P_m x\rangle = \lim_{m\to\infty}\langle p, (I-P_m)x\rangle &= \lim_{m\to\infty}\langle (I-P_m^*)p, x\rangle \\ &= \lim_{m\to\infty}\lim_{j\to\infty}\langle (I-P_m^*)p, x_j\rangle \\ &= \lim_{j\to\infty}\lim_{m\to\infty}\langle (I-P_m^*)p, x_j\rangle = 0,\end{aligned}$$

where the forth equality exploits the commutativity of the double limit of the double sequence (see [2, Theorem 9.16]). Therefore, $P_m x \to x$ weakly.

(ii): Let $p \in E^*$ be arbitrarily fixed. It suffices to show that there is a subsequence $\{Q_m p\}_{m\in\mathbb{N}}$ of $\{Q_n p\}_{n\in\mathbb{N}}$ such that $\langle Q_m p, x\rangle \to \langle p, x\rangle$ for every $x \in E$. By Proposition 6.1(ii), there exists $p_n \in W^n$ such that $p_n \to p$ weakly*. Choose any $x \in E$. Since $\{(I-Q_n^*)x\}_{n\in\mathbb{N}}$ is a bounded sequence in E^{**} in view of $\mathcal{R}(I-Q_{n+1}^*) \subset \mathcal{R}(I-Q_n^*) \subset \cdots$ for each $n \in \mathbb{N}$ and $\{p_i\}_{i\in\mathbb{N}}$ is a bounded sequence in E^*, the numerical double sequence $\{\langle p_i, (I-Q_n^*)x\rangle\}_{(i,n)\in\mathbb{N}^2}$ contains a convergent subsequence $\{\langle p_j, (I-Q_m^*)x\rangle\}_{(j,m)\in\mathbb{N}^2}$. It follows from $I - Q_m^* = (W^m)^\perp$ and $p_j \in W^j$ that $\langle p_j, (I-Q_m^*)x\rangle = 0$ for each $j, m \in \mathbb{N}$ with $j \le m$. Hence,

$$\begin{aligned}\lim_{m\to\infty}\sup_{j\in\mathbb{N}} |\langle p_j, (I-Q_m^*)x\rangle| &= \lim_{m\to\infty}\sup_{j\ge m} |\langle p_j, (I-Q_m^*)x\rangle| \\ &= \lim_{j,m\to\infty} |\langle p_j, (I-Q_m^*)x\rangle|.\end{aligned}$$

32

Figure 3: Printed page 32 of the source, containing Theorem 6.3(ii) and the beginning of its proof.

2 Idea of the proof

Positivity is not a formal consequence of idempotence: an oblique projection can send a positive vector outside the induced positive cone of its range. This already happens in two dimensions.

Likewise, density of the ranges of finite-rank projections does not control the norms of the projections. On ℓ^2, start with the usual coordinate truncation and add an increasingly large multiple of the next coordinate in the e_1-direction. The added term vanishes on the range, so idempotence is preserved, while a single square-summable vector makes the first coordinate of every projected vector diverge. Weak convergence is then impossible.

3 Counterexample to Theorem 6.2

Theorem 1. *There is an ordered finite-dimensional Banach space E, a closed subspace V, and a continuous projection $P : E \to V$ which is not positive. Indeed, V can be chosen so that no positive projection of E onto V exists.*

Proof. Let $E = \mathbb{R}^2$, with its usual norm and coordinatewise order, so $E_+ = \mathbb{R}^2_+$. Put

$$V = \operatorname{span}\{(1, -1)\}, \qquad P(a, b) = (a, -a).$$

Then P is linear and continuous, its range is V, and

$$P^2(a, b) = P(a, -a) = (a, -a) = P(a, b),$$

so it is a continuous projection onto the closed subspace V. However, $(1, 0) \in E_+$, whereas

$$P(1, 0) = (1, -1) \notin E_+ \cap V = V_+.$$

Thus P is not positive.

Moreover $V_+ = V \cap E_+ = \{0\}$. If a positive projection $Q : E \to V$ existed, then $Q(E_+) \subseteq V_+ = \{0\}$. Since $E = E_+ - E_+$, linearity would force $Q = 0$, contradicting that its range is the nonzero space V. Hence this subspace admits no positive projection at all. □

There is also an immediate internal inconsistency in the printed separation argument. It asserts the existence of x^* such that $\langle Px, x^* \rangle = 1$ and $\langle y, x^* \rangle = 0$ for every $y \in V$. But $Px \in V$, so the second equality applied to $y = Px$ already gives $\langle Px, x^* \rangle = 0$.

4 Counterexample to Theorem 6.3

Theorem 2. *Both parts of Theorem 6.3 are false under their displayed hypotheses. More precisely, there are nested finite-dimensional subspaces with dense union and continuous projections onto them for which one fixed vector has no subsequence of projections converging weakly to that vector.*

Proof. Let $E = \ell^2$ with standard orthonormal basis $(e_k)_{k\geq 1}$, and let

$$V_n = \operatorname{span}\{e_1, \ldots, e_n\}.$$

Then $V_1 \subset V_2 \subset \cdots$, and the union is norm dense in ℓ^2. For $x = (x_k)_{k\geq 1}$, define

$$P_n x = \sum_{k=1}^{n} x_k e_k + n^3 x_{n+1} e_1. \tag{1}$$

Each P_n is bounded, since

$$\|P_n x\|_2 \le \left\| \sum_{k=1}^{n} x_k e_k \right\|_2 + n^3 |x_{n+1}| \le (1+n^3)\|x\|_2.$$

Its range is contained in V_n, and if $y \in V_n$, then $y_{n+1} = 0$, so $P_n y = y$. Consequently $P_n^2 = P_n$ and $\mathcal{R}(P_n) = V_n$: it is a continuous projection onto the required Galerkin subspace.

Now take

$$x_1 = 0, \qquad x_k = \frac{1}{k^2} \quad (k \ge 2).$$

This vector belongs to ℓ^2. Evaluation against e_1, a continuous linear functional on ℓ^2, gives

$$\langle P_n x, e_1 \rangle = x_1 + n^3 x_{n+1} = \frac{n^3}{(n+1)^2} \longrightarrow +\infty.$$

Every subsequence has indices tending to infinity, so no subsequence of $(P_n x)$ can converge weakly to x (or weakly to any vector). This refutes part (i).

For part (ii), identify $E^* = \ell^2$ in the usual self-dual way, set $W_n = V_n$, and set $Q_n = P_n$. Norm density implies weak-star density, so (W_n) is a Galerkin approximation scheme for (E^*, w^*). Taking $p = x$, the same pairing with $e_1 \in E$ diverges. Hence no subsequence of $(Q_n p)$ converges weak-star to p, refuting the intended assertion of part (ii). □

5 Where the printed proof fails and a repair

The source proof says that $((I - P_n^*)p)$ is bounded "in view of" an inclusion of ranges. The displayed assumptions impose neither compatibility of the projections nor uniform boundedness of their operator norms; the counterexample has $\|P_n\| \ge n^3$. Mere inclusion of finite-dimensional ranges cannot supply the missing norm estimate.

A standard sufficient repair for part (i) is

$$\sup_n \|P_n\| < \infty. \tag{2}$$

Indeed, write $M = \sup_n \|P_n\|$. Given $x \in E$ and $\varepsilon > 0$, choose m and $y \in V_m$ with $\|x - y\| < \varepsilon/(M+1)$. For every $n \ge m$, nestedness gives $y \in V_n$, hence $P_n y = y$, and

$$\|P_n x - x\| \le \|P_n(x-y)\| + \|y - x\| < \varepsilon.$$

Thus (2) yields the stronger conclusion $P_n x \to x$ in norm. Positivity, however, requires an independent order-theoretic hypothesis; it does not follow from continuity and idempotence.

6 Impact on the equilibrium arguments

In Step 1 of the proof of Theorem 7.1, the source explicitly invokes Theorem 6.2 to assert that $P_n : E \to V_n$ is positive and hence that $P_n(X(t)) \subset V_{n,+}$. In Step 2 it invokes Theorem 6.3(i) to identify weak limits of the projected endowments. The proof of Theorem 7.2 repeats the same scheme in the weak-star setting. The counterexamples above show that these two inputs are unavailable under the hypotheses stated in Section 6.

There is a further definitional issue in Step 1: the source sets $X^n(t) = P_n(X(t))$ and calls $\succsim_n (t)$ the restriction of a preference relation originally defined only on $X(t)$ to $X^n(t)$. For a general projection, $P_n(X(t))$ need not be contained in $X(t)$, so such a restriction need not be defined.

These observations establish a proof gap, not a counterexample to Theorems 7.1 or 7.2. Other equilibrium-existence methods or stronger approximation hypotheses may repair the main conclusions. They also do not decide the concluding question asking whether saturation is necessary.

7 Verification and novelty check

The proof is algebraic. The accompanying script `code/check_finite_truncations.py` checks, for finite test vectors and many values of n, that (1) is idempotent on its finite-coordinate model, fixes V_n, and has the predicted divergent first coordinate. These checks are sanity tests, not a substitute for the proof.

A bounded novelty search on 27 August 2026 covered: the arXiv corpus and the run's local indexes for the paper id, title, "continuous projection," "positive linear operator," "Theorem 6.2," and "Theorem 6.3"; web searches for the exact theorem phrase, the title together with "erratum" or "projection positive," and the authors together with "Galerkin projection." The search found the original arXiv record and a 2017 talk repeating the positivity step, but no erratum or published correction. This is evidence only within the stated search bounds, not a claim of exhaustive novelty.

8 Human-review recommendation

1. Verify that the published version has the same hypotheses as arXiv v2, especially whether compatibility or uniform boundedness of the projections was added.
2. Check whether Theorems 7.1 and 7.2 have independent proofs in the cited equilibrium literature; the present result concerns the proof printed here.
3. If communicating the issue to the authors, lead with the two-line $\mathbb{R}^2$ counterexample and the explicit ℓ^2 family (1).

References

[1] M. Ali Khan and N. Sagara, *Fatou's Lemma, Galerkin Approximations and the Existence of Walrasian Equilibria in Infinite Dimensions*, Pure Appl. Funct. Anal. 2 (2017), 317–355; arXiv:1610.04776v2.